

Zaid Helsinki Putra

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Kendari, Southeast Sulawesi, Indonesia

Bachelor of Science (S1) Informatics Engineering student at Universitas Halu Oleo (GPA 3.91/4.00) with a strong focus on **Artificial Intelligence, Machine Learning, and Data Platform Engineering**. Experienced building end-to-end, data-driven systems from geospatial and NLP data pipelines to deep learning models and full-stack web/database platforms including flood-risk mapping, a RAG-based health chatbot, a GIS portal serving international research partners, and a government case-reporting system. Recipient of multiple national research grants (PKM) and a speaking invitation at the 5th International Conference on Green Sustainable Technology and Management (Malaysia). Now seeking an **AI & Data Platform Intern** role to apply and grow these skills in a production data environment.

EDUCATION

Universitas Halu Oleo - Kendari, Indonesia
Bachelor of Engineering, Informatics Engineering - GPA 3.91/4.00

Sep 2023 – Present

KEY PROJECTS

WebGIS Desa Bajo - Spatial Mapping of Education Facilities

2026

Partnership with PUSKAPA UI & Macquarie University

- Standardized education-facility spatial data into GeoJSON/WMS and built a Laravel + MySQL backend connected to an interactive Leaflet.js map for use by academic partners and the public.

Tech: Laravel, PostgreSQL/PostGIS, Leaflet.js, QGIS, Bootstrap - Live: geoportal-desabajo.my.id

RAG Chatbot for Maternal & Child Health / Stunting Risk Education

2026

Course Project - Deep Learning

- Built a structured knowledge base from health-department documents and implemented semantic search with Sentence-BERT/IndoBERT embeddings to retrieve relevant context.
- Designed a Retrieval-Augmented Generation architecture connected to an LLM to generate factual, hallucination-resistant chatbot responses.

Tech: Python, IndoBERT, Sentence-BERT, LangChain/LlamaIndex, Vector DB (Chroma/FAISS), LLM API

Cacao Leaf Nutrient Deficiency Detection via Transfer Learning

2026

Course Project - Deep Learning

- Built and augmented a labeled cacao-leaf image dataset, then compared Vision Transformer, Swin Transformer, and MobileViT transfer-learning approaches, deploying the best model to a Streamlit web app for real-time detection.

Tech: Python, PyTorch, ViT, Swin Transformer, MobileViT, Streamlit

Land Cover Classification & NDVI Vegetation Trend Analysis (2020–2025)

2025

Course Project - Digital Image Processing

- Processed and cloud-masked five years of Sentinel-2 imagery in Google Earth Engine, computed NDVI indices, and ran supervised Random Forest classification to map land-cover change in Kendari Barat.

Tech: Google Earth Engine, Python/JavaScript API, Sentinel-2 Imagery, Random Forest

Public-Space Violence Risk Prediction for Women

2025

Course Project - Data Science

- Handled high-percentage missing values and optimized categorical encoding for high-cardinality features, then tuned a LightGBM model with Optuna to predict location-level safety risk.

Tech: Python, LightGBM, Optuna, Target Encoding, Pandas, Scikit-Learn

Typo-Tolerant Information Retrieval for Local News Search

2026

Course Project - Information Retrieval

- Scraped and preprocessed Southeast Sulawesi local news articles, then implemented fuzzy/edit-distance matching within a TF-IDF/inverted-index search structure, evaluating results with Precision, Recall, and MAP across typo-variant queries.

Tech: Python, NLTK/Sastrawi, Levenshtein Distance, TF-IDF, Inverted Index

WORK EXPERIENCE

Dinas Pemberdayaan Perempuan dan Perlindungan Anak (DP3A) - Kendari

Jan 2026 – Mar 2026

Web Developer & System Analyst (Internship)

- Designed the system workflow (UML/BPMN) and a NoSQL database schema for a case-reporting system for UPTD PPA; built RESTful APIs with Node.js and Express handling sensitive-data encryption and case-status management, with a React.js front end for public reporting and admin monitoring.
- Built a web-based Employee Information System (Sistem Informasi Kepegawaian) for the department.
- Designed the department's new logo as part of a visual identity refresh.

Tech: Node.js, Express, MongoDB, React.js

SMK Negeri 4 Kendari - Kendari

Sep 2025 – Oct 2025

Web Developer & System Analyst (Thematic Community Service)

- Conducted system analysis to identify administrative-workflow requirements, then assisted in designing and developing web-based solutions supporting school operations and information management, collaborating with teachers and staff through testing and go-live.
- Developed and delivered a Microsoft Word training curriculum and hands-on materials for students as part of a related digital office-skills program.

Universitas Halu Oleo – Kendari

Jul 2026

Liaison Officer, ASIIN International Accreditation Visitation

- Served as the main point of contact for the international ASIIN assessor; coordinated logistics, accommodation, and scheduling, and facilitated bilingual (English–Indonesian) communication with department and faculty leadership.

AWARDS & HONORS

- Funding Recipient, PKM-KI (Karya Inovatif) – Student Creativity Program (PKM) 2026, Kemendikbudristek (Team Lead)
- 1st Place, Marine Transportation Sub-Theme – National Scientific Paper Competition (LKTIN) 2026, Faculty of Engineering, Halu Oleo University – “Digital Twin Maritime Transport System Based on AI and CVRP”
- Speaker, 5th International Conference on Green Sustainable Technology and Management (ICGSTM) 2026, INTI International University, Malaysia – “A MobileNetV3 Deep Learning Approach for Climate-Resilient Post-Harvest Coffee Quality Classification”
- Best Poster & Certificate of Poster Presenter, ICGSTM 2026 – “Development of an AI-Driven Dehumidification Drying System for Preserving Patchouli Alcohol”
- Best Full Paper Award, AFC Season 17 Scientific Paper Competition 2026, Faculty of Agriculture, Jenderal Soedirman University – “Coffee Tech Ecosystem: IoT and AI Optimization for Local Coffee Processing”
- 1st Place, Business Plan Competition (IT FEST) 2025, Dept. of Informatics Engineering, Halu Oleo University – “meRangkul” adaptive digital therapy platform
- 1st Place, Functional Writing Competition (Natural Resources Category) – FORKESTRA 2024, Bank Indonesia Representative Office of Southeast Sulawesi
- PKM-GFT Incentive Awardee 2025 – “CereBros” Brain-Computer Interface + AI + Holography communication system, Kemendikbudristek
- PKM-GFT Incentive Awardee 2024 – “AquaRevive Center” AI + IoT clean-water management system, Kemendikbudristek
- Book Author, “Office Mastery: Training Guide for Microsoft Word, Excel & PowerPoint” (ISBN 978-634-7362-30-8), Arsy Media Publisher, 2025
- Sultra Cerdas Scholarship Awardee, 2025

PUBLICATIONS

- “Pendampingan Implementasi Sistem Informasi Pelaporan dan Pemetaan Kasus Kekerasan Berbasis Web dan GIS pada DP3A Kota Kendari” – Community Service (Pengabdian), Sinta 5, Published 2026. DOI: 10.54082/jamsi.3149
- “Optimizing Typo-Tolerant Information Retrieval for Southeast Sulawesi Local News Search” – Research Paper, Sinta 5, Under Review, 2026
- “Pemberdayaan UMKM Melalui Pelatihan Membangun Hubungan Pelanggan Berbasis WhatsApp Business” – Community Service (Pengabdian), Sinta 4, Published 2026. DOI: 10.58266/jpmb.v4i4.1535

CERTIFICATIONS & TRAINING

- Belajar Dasar AI - Dicoding, 2026
- Belajar Machine Learning untuk Pemula – Dicoding, 2026
- Memulai Pemrograman dengan Python – Dicoding, 2026
- Belajar Penerapan Data Science dengan Microsoft Fabric – Dicoding, 2026
- Membangun Aplikasi Gen AI dengan Microsoft Azure – Dicoding, 2026
- Prompt Engineering untuk Software Developer – Dicoding, 2026
- Introduction to Financial Literacy – Dicoding, 2026
- Data Science with Microsoft Fabric & Gen AI with Azure – Microsoft Elevate Training, 2026

ORGANIZATION EXPERIENCE

Kawan Inspirasi Kendari – Kendari

Dec 2024 – Present

Partnership Division Member (Co-Coordinator)

A non-profit organization dedicated to children's education in underserved coastal and inland communities.

- Map and pitch prospective sponsors, government partners, and communities; negotiate and maintain partnerships supporting program sustainability.
- Coordinated logistics for two multi-day outreach programs (Sapa Pulau, Wakatobi & Bau-Bau) as committee member and secretary.

SKILLS

- **Languages:** Indonesian (native), English (Intermediate – B1)
- **AI/ML:** Machine Learning, Deep Learning (CNN, ViT/Swin/MobileViT, Autoencoders), NLP (IndoBERT, Sentence-BERT), RAG & LLM Integration (LangChain, LlamaIndex), Scikit-Learn, XGBoost/LightGBM/CatBoost, Optuna, PyTorch/TensorFlow
- **Data & Platform Engineering:** Python, SQL, MongoDB, PostgreSQL/PostGIS, RESTful API Design (Node.js/Express), Vector Databases (Chroma/FAISS), Google Earth Engine, Microsoft Fabric & Azure
- **Web Development:** Laravel, React.js, Node.js/Express, Leaflet.js, Bootstrap, PHP
- **GIS / Geospatial:** QGIS, GeoPandas, PostGIS, Google Earth Engine, Remote Sensing (Sentinel-2)
- **Soft Skills:** Public Speaking, Research & Scientific Writing, Cross-Functional Collaboration, Time Management, Problem Solving